IILM UNIVERSITY, GREATER NOIDA		LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET		
EXPERIMENT TITLE: Study of both the current - voltage characteristics and the power curve to find the maximum power point (MPP) and efficiency of a solar cell.		
EXPERIMENT NO. 02	DEPT. : Physics	ISSUE DATE: 12 August, 2026
LABORATORY: 26SOSP101 Semiconductor and Quantum Physics Lab	SEMESTER: I, II	TOTAL PAGE: 5

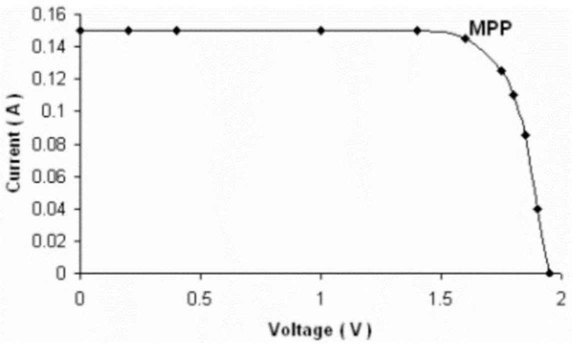
AIM: Study of both the current - voltage characteristics and the power curve to find the maximum power point (MPP) and efficiency of a solar cell.

APPARATUS: Experimentation with Solar Energy Nvis 6005, Solar Panel, DB15 connector, Patch cords

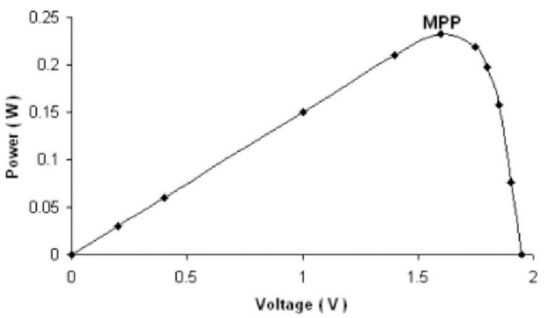
THEORY AND FORMULA USED: The maximum power point (MPP) of a solar cell is the point in the V-I characteristics, at which the product of voltage and current is greatest.

Otherwise, the maximum power point (MPP) is the maximum value of power in the Voltage-Power curve. The resistance, R_{MPP}, at which the output power is at a maximum can be calculated using the following formula:

$$R_{MPP} = \frac{V_{MPP}}{I_{MPP}}$$



Current-voltage characteristic of the solar cell



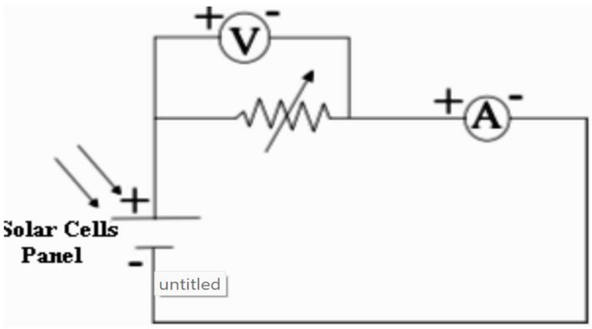
Power curve of the solar cell as a function of voltage

PROCEDURE:

1. Take the Experimentation with Solar Energy Nvis 6005 along with solar panel.
2. Place the solar panel in the stand and adjust the panel at an angle about 45^0 with the ground. Direct the sunlight straight at the solar panel (angle of 90^0).

   Note: If sunlight is not properly available then any source of light like lamp can be used.

3. With the DB15 connector, connect the Experimentation with Solar Energy Nvis 6005 with the solar panel. Then wait for 1 minute to avoid errors due to temperature fluctuations.
4. Set the potentiometer to maximum resistance i.e. at fully clockwise position and measure and record its resistance into the observation table.
5. Connect the solar cell as shown in the following circuit diagram.



Setup for determining characteristics of a solar cell

   - Connect positive terminal of solar cell to P1 terminal of the potentiometer.
   - Connect the other end of potentiometer i.e., P2 to positive terminal of ammeter.
   - Connect negative terminal of ammeter to negative terminal of solar cell.
   - Now connect the positive terminal of voltmeter to P1 and negative terminal of voltmeter to P2.

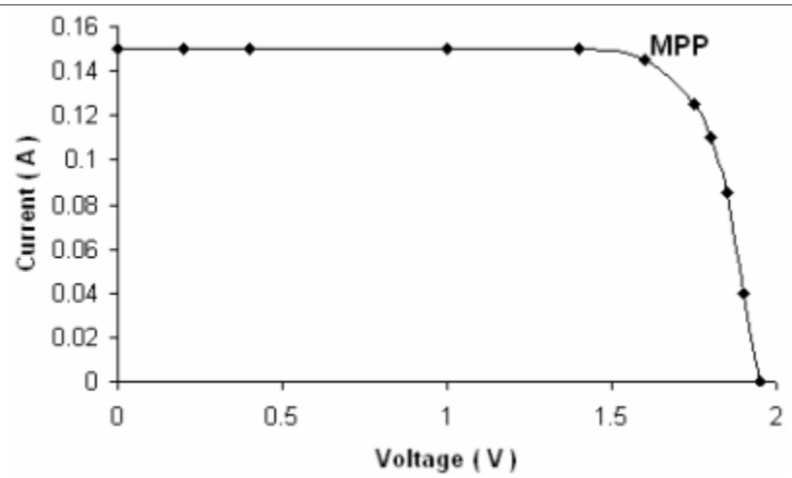
6. Record the values of corresponding voltage and current into the observation table.
7. Now gradually move the potentiometer in anti-clockwise direction so that the resistance of the potentiometer decreases. Now measure the resistances at successively smaller values and record the corresponding values of voltages and current into the observation table below.

Note: Always to measure the resistance of potentiometer at any position, first remove the patch cords from P1 and P2 and measure resistance by multimeter. Reconnect these connections again for further measurements**.**

OBSERVATION TABLE:

Sr No.	**Resistance, R (Ω)**	**Voltage, V (Volts)**	**Current, I (mA)**	**Power calculated P= V.I (Watts)**
1				
2				
3				
4				
5				
6				
7				

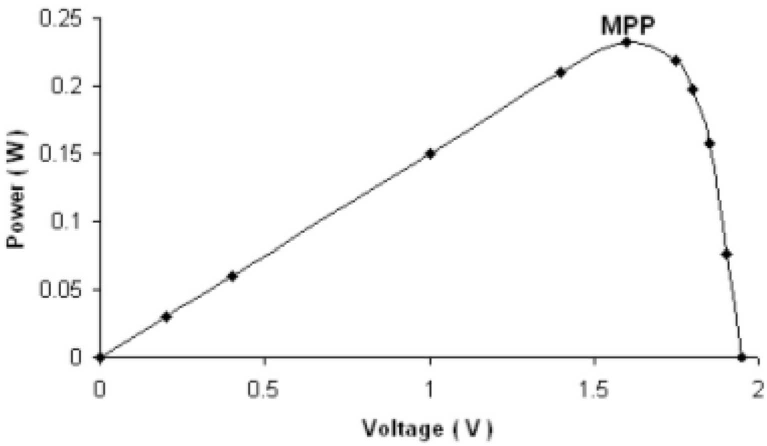
8. Plot the I-V characteristics from the measurements recorded in the table, to show how the photoelectric current depends on the photoelectric voltage and to find maximum power point. Expected I-V curve is as follows



Current-voltage characteristic of the solar cell

From V-I characteristics you can easily find the maximum power point (MPP). MPP occurs where the product of voltage and current is greatest. Plot the curve of power as a function of voltage from the measurements recorded in the table.

Expected Power-Voltage curve is as follows



Power curve of the solar cell as a function of voltage

The maximum power point (MPP) is the maximum value of power in the above curve.

The resistance, RMPP, at which the output power is at a maximum can be calculated using the following formula:

$$R_{MPP} = \frac{V_{MPP}}{I_{MPP}}$$

RESULT:

PRECAUTIONS:

1. Avoid short circuits; connect wires properly and check for loose connections.
2. For accuracy, take multiple readings and calculate the average.
3. Connect the solar cell to the circuit with the correct polarity (positive to positive, negative to negative).
4. Use a calibrated voltmeter and ammeter suitable for the expected voltage/current range.

VIVA QUESTIONS:

1. What is a solar cell?
2. What is the working principle of a solar cell?
3. Which material is commonly used to make solar cells?
4. What is the difference between a solar cell and a solar panel?